PROJECT

Sonar: Mines vs Rocks

Logistic regression with standard scaling

Classify a sonar return as a metal cylinder (mine) or a rock.

TEAM 55 · MEMBERS & ROLES

Arnav Yadav
2026EEB1326
Documentation lead
Data & exploration
Load and inspect the dataset; create class and signal summaries.

Dhiraj Kumar Yadav
2026MCB1525
Execution lead
Preprocessing
Scale features; rank correlations; select the top 20.

Rishabh Mahar
2026EEB1399
PPT lead
Models & evaluation
Train both logistic models and compare test metrics.

Kashish Chhabra
2026AIB1048
Video lead
Plots & results
Create evaluation charts, confusion matrix and results table.

Team number: 55

GitHub repository URL: https://github.com/arnav-7003/AI-Project

Problem and data

THE QUESTION

Can a sonar echo be classified as a mine or a rock?

Each echo is represented by 60 measurements of energy across frequency bands. The model predicts the object class from that signal.

TARGET LABELS

M = mine (metal cylinder)
R = rock

DATASET AT A GLANCE

208
labelled sonar patterns

60
numeric features

111 mines

97 rocks

Values range from 0 to 1; the dataset has no missing values.

DATA SOURCE

UCI Machine Learning Repository
Connectionist Bench (Sonar, Mines vs. Rocks)

Approach

01

Load

208 rows
60 bands + M/R label

→

02

Split

Stratified 80/20
seed = 42

→

03

Scale

StandardScaler
fit on train only

→

04

Compare

Logistic regression
all 60 vs top 20
(correlation-ranked)

→

05

Evaluate

Held-out test set
accuracy · precision
recall · F1

LIBRARIES

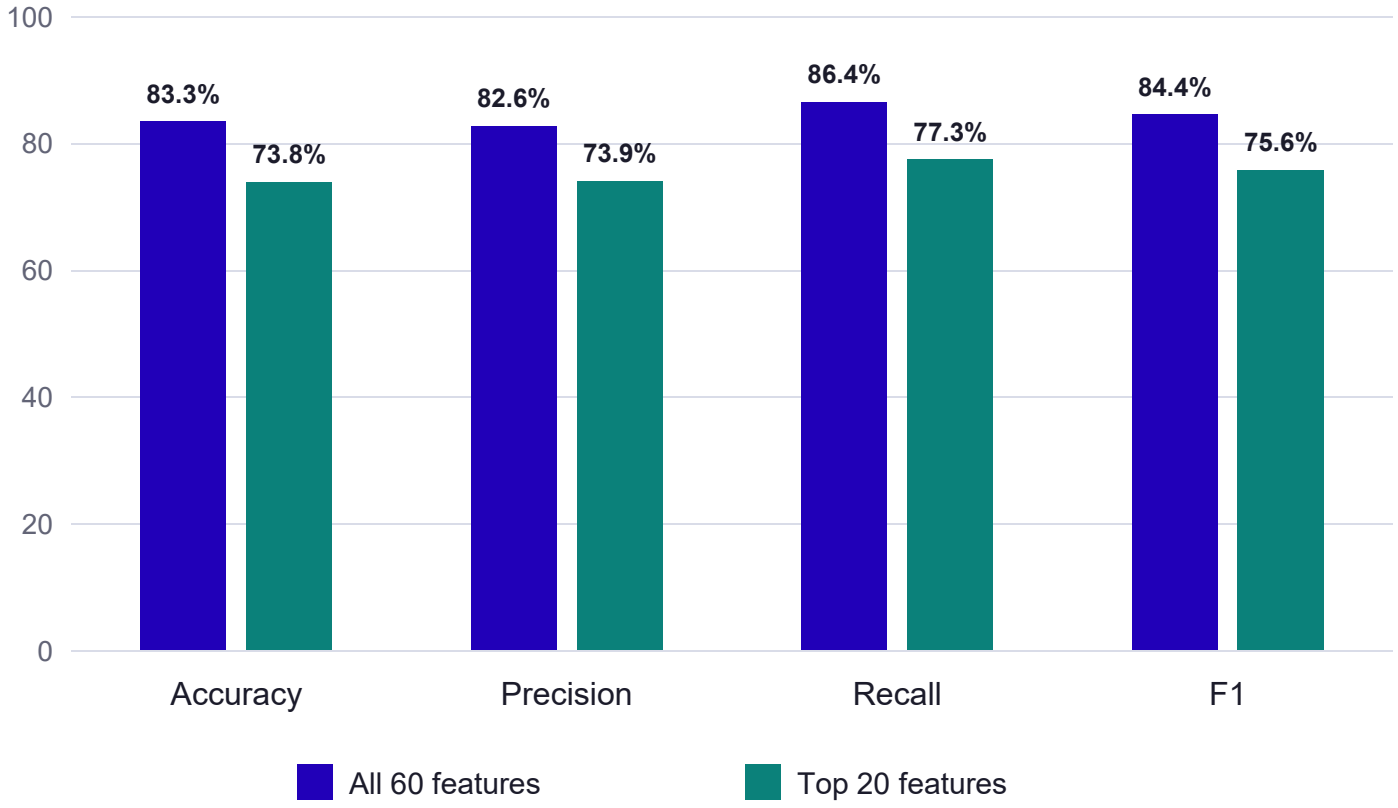
Python · pandas · NumPy · scikit-learn · Matplotlib

The scaler and feature ranking use training data only.

Results on one held-out split

All 60 features scored 83.3% accuracy; selecting 20 features reduced accuracy by 9.5 points.

TEST METRICS (%)



EVALUATION OUTPUT

Evaluation output · fixed 80/20 split

Model	Accuracy
All 60 features	83.3%
Top 20 features	73.8%

ALL 60 FEATURES · ACTUAL × PREDICTED

Actual / predicted	Rock	Mine
Rock	16	4
Mine	3	19

Train: 166 · Test: 42 · Correct: 35/42

Learnings and contributions

WHAT WORKED / WHAT DIDN'T

Worked: scaling and feature ranking use training data only.

Didn't: the top-20 model scored 9.5 points lower in accuracy than all 60 features on this split.

Limit: one dataset and one test split leave repeatability uncertain.

CONTRIBUTIONS IN main.py

Arnav Yadav	Data loading and exploration
Dhiraj Kumar Yadav	Preprocessing and feature selection
Rishabh Mahar	Training and evaluation
Kashish Chhabra	Plots and results table

AI-USE DISCLOSURE

OpenAI Codex helped summarize the supplied code and draft this deck. Metric values were independently recomputed from the provided dataset using the code's fixed split and logistic model objective; run the project with its dependencies to verify before submission.